VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB RECORD

Bio Inspired Systems (23CS5BSBIS)

Submitted by

SANTHOSH N (1BM23CS302)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
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B.M.S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled " Bio Inspired Systems (23CS5BSBIS)" carried out by **SANTHOSH N (1BM23CS302),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements of the above mentioned subject and the work prescribed for the said degree.

Rohith Vaidya K Assistant Professor Department of CSE, BMSCE	Dr. Kavitha Sooda Professor & HOD Department of CSE, BMSCE

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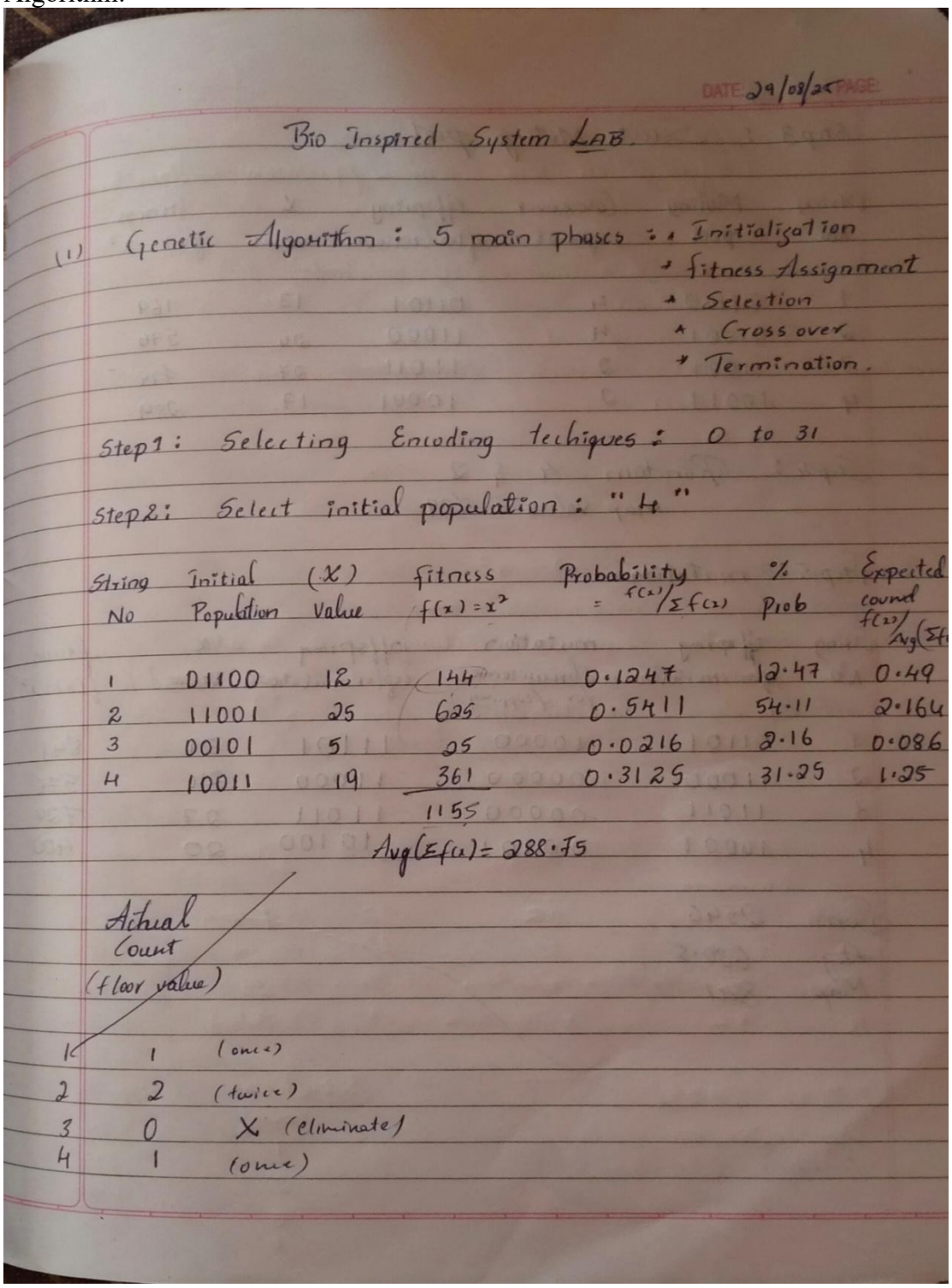
Github Link:

https://github.com/SANTHoshN302/BIS_302

Program 1 : Genetic Algorithm for Optimization Problems:

Algorithm:



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Bio Inspired System LAB.

(1) Genetic Algorithm : 5 main phases :
- Initialization
- fitness Assignment
- Selection
- Cross over
- Termination.

Step1 : Selecting Encoding techiques : 0 to 31

Step2 : Select initial population : "4"

String No	Initial Population	(X) Value	fitness $f(x)=x^2$	Probability $f(x)/\Sigma f(x)$	% Prob	Expected count $f(x)/Avg(\Sigma f$
1	01100	12	144	0.1247	12.47	0.49
2	11001	25	625	0.5411	54.11	2.164
3	00101	5	25	0.0216	2.16	0.086
4	10011	19	361	0.3125	31.25	1.25
			1155			

Avg(Σf(x)) = 288.75

	Actual Count (floor value)	
1	1	(once)
2	2	(twice)
3	0	X (eliminate)
4	1	(once)

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Step 3 : Select Mating pool.

String No	Mating pool	Crossover point	Offspring after Crossover	x value	fitness $f(x) = x^2$
1	01100	4	01101	13	169
2	11001	4	11000	24	576
3	11001	2	11011	27	729
4	10011	2	10001	17	289

Step 4 : Random 4 & 2
map value = 729

Step 5 : mutation.

String No	offspring after crossover	mutation Chromosome for flipping	offspring after mutation	x (value)	fitness $f(x) = x^2$
1	01101	10000	11101	29	841
2	11000	00000	11000	24	576
3	11011	00000	11011	27	729
4	10001	00101	10100	20	400

Sum 2546
Avg 630.5
Map 841

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Output:

Best fitness

Pseudo Code :

Step 1: Set population size, maximum Generation, mutation rate and chromosome length.

Step 2: Calculate probability, Expected count and Actual count for initial population.

Step 3: Perform Crossover on initial population to get new offspring set.

Step 4: Perform mutation over obtained offspring set and get maximum fitness and corresponding 'x' value.

Step 5: repeat the iterations to get best feasible fitness and 'x' value and print the same.

Output :

Best fitness : 18
Best fitness : 19
Best fitness : 18
Best fitness : 18
Best fitness : 19
Best fitness : 20
Best fitness : 19
Best fitness : 20
:
Best fitness : 20

Best Solution found :
fitness : 20

29/8/25

Code:

```
import random
import math

def fitness_function(x):

    return x * math.sin(10 * math.pi * x) + 1

POPULATION_SIZE = 20
MUTATION_RATE = 0.1
CROSSOVER_RATE = 0.8
GENERATIONS = 100
X_BOUND = (-1, 2)  # Search space for x

def create_individual():
    return random.uniform(*X_BOUND)

def create_population(size):
    return [create_individual() for _ in range(size)]

def evaluate_population(population):
    return [fitness_function(ind) for ind in population]

def select(population, fitnesses):
    total_fitness = sum(fitnesses)
    probs = [f / total_fitness for f in fitnesses]
    return random.choices(population, weights=probs, k=2)

def crossover(parent1, parent2):
    if random.random() < CROSSOVER_RATE:
        alpha = random.random()
        child1 = alpha * parent1 + (1 - alpha) * parent2
        child2 = alpha * parent2 + (1 - alpha) * parent1
        return child1, child2
    else:
        return parent1, parent2


def mutate(individual):
    if random.random() < MUTATION_RATE:
        mutation = random.uniform(-0.1, 0.1)
        individual += mutation
        individual = max(min(individual, X_BOUND[1]), X_BOUND[0])  # Keep in bounds
    return individual

def genetic_algorithm():
    population = create_population(POPULATION_SIZE)
    best_individual = None
    best_fitness = float('-inf')
```

```
    for generation in range(GENERATIONS):
        fitnesses = evaluate_population(population)

            for i, fit in enumerate(fitnesses):
             if fit > best_fitness:
                best_fitness = fit
                best_individual = population[i]

        new_population = []
        while len(new_population) < POPULATION_SIZE:
            parent1, parent2 = select(population, fitnesses)
            child1, child2 = crossover(parent1, parent2)
            child1 = mutate(child1)
            child2 = mutate(child2)
            new_population.extend([child1, child2])

        population = new_population[:POPULATION_SIZE]

        print(f"Generation {generation+1}: Best Fitness = {best_fitness:.5f}, Best X =
{best_individual:.5f}")

    print("\nOptimization complete!")
    print(f"Best solution: x = {best_individual:.5f}, f(x) = {best_fitness:.5f}")

if __name__ == "__main__":
    genetic_algorithm()
```

Output:

```
= RESTART: C:/Users/student/AppData/Local/Progr
Best Fitness = 16
Best Fitness = 17
Best Fitness = 17
Best Fitness = 18
Best Fitness = 19
Best Fitness = 18
Best Fitness = 18
Best Fitness = 18
Best Fitness = 19
Best Fitness = 20
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Best Fitness = 20

Best solution found:
Fitness (X value): 20
```
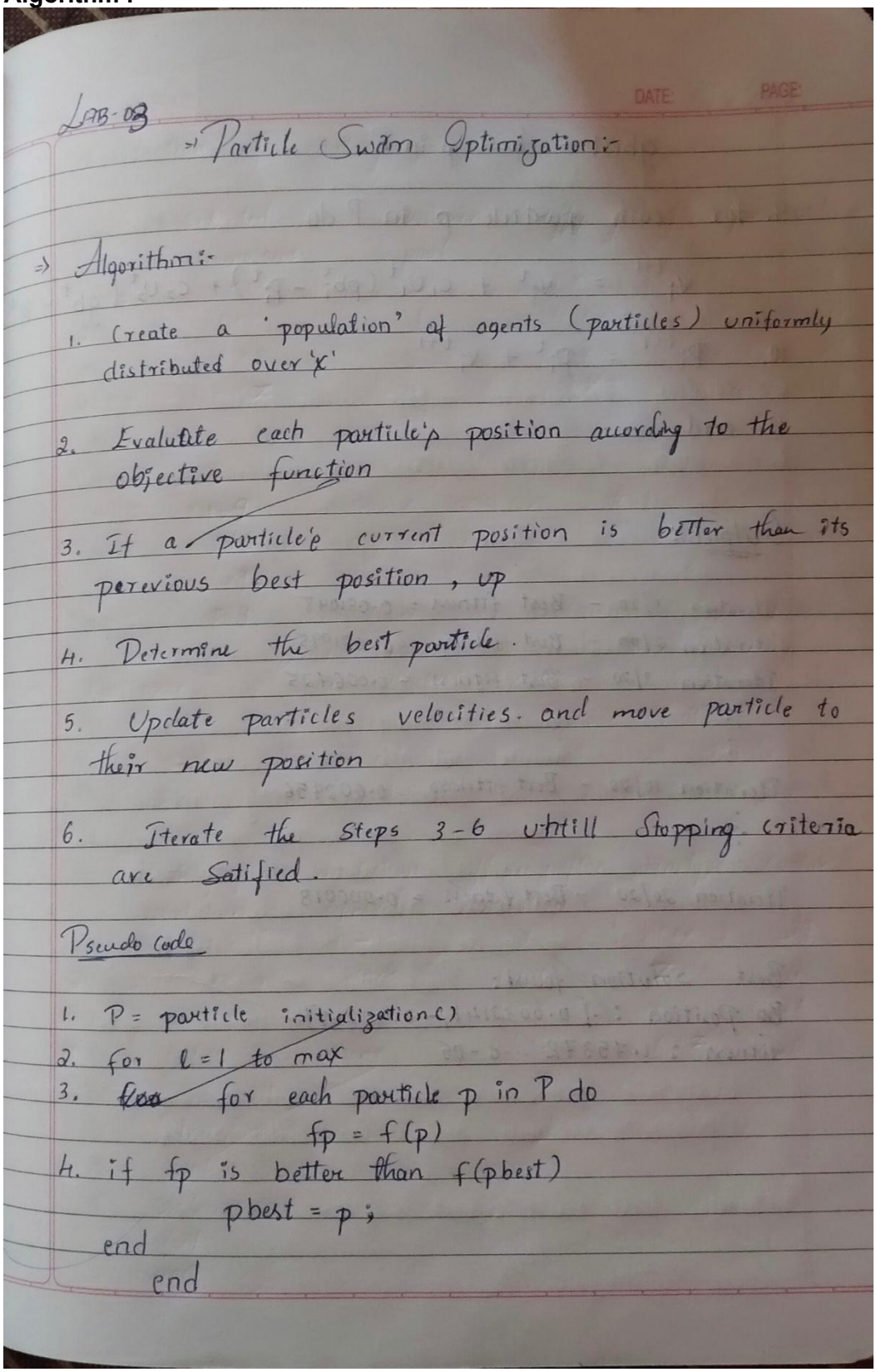

Program 02: **Particle Swarm Optimization for Function Optimization:**

Algorithm :



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=> Particle Swarm Optimization:-

=> Algorithm:-

1. Create a 'population' of agents (particles) uniformly distributed over 'x'
2. Evaluate each particle's position according to the objective function
3. If a particle's current position is better than its perevious best position, up
4. Determine the best particle.
5. Update particles velocities. and move particle to their new position
6. Iterate the steps 3-6 uhtill Stopping criteria are Satified.

Pseudo Code

```
1. P = particle initialization()
2. for l = 1 to max
3.     for each particle p in P do
           fp = f(p)
4. if fp is better than f(pbest)
       pbest = p;
   end
end
```

gbest = best p in P

9. for each particle p in P do

$$v_i^{t+1} = v_i^t + c_1 U_1^t (pb_i^t - p_i^t) + c_2 U_2^t (gb^t - p_i^t)$$

10. $p_i^{t+1} = p_i^t + v_i^{t+1}$

11. end.

Output :

Iteration 1/20 - Best fitness = 0.081095
Iteration 2/20 - Best fitness = 0.081095
Iteration 3/20 - Best fitness = 0.006725
:
:
Iteration 10/20 - Best fitness = 0.002956
:
:
Iteration 20/20 - Best fitness = 0.000018

Best solution found:
Ba Position : [0.0032141, -0.00268]
fitness : 1.75372 ... e-05

Iteration 1.

Particle No	initial Position x	y	velocity x	y	Best x	y	fitness value.
P1	1	1	0	0	-	-	2
P2	-1	+1	0	0	-	-	2
P3	0.5	-0.5	0	0	-	-	0.5
P4	1	-1	0	0	-	-	2
P5	0.25	-0.25	0	0	-	-	0.25

Iteration 2.

Pno	initial x	y	velocity x	y	Best Sol	Best Pos x	y	fitness.
P1	1	1	-0.75	-0.75	2	1	-1	2
P2	-1	1	-1.25	1.25	2	-1	1	2
P3	0.5	-0.5	-0.25	0.75	0.5	0.5	0.5	0.5
P4	1	-1	-0.75	1.25	2	1	-1	2
P5	0.25	0.25	0	0	0.125	0.25	0.25	0.125

Iteration 3.

Pno	initial x	y	velocity x	y	Best Sol	Best Pos x	y	fitness.
P1	0.25	0.25	-0.375	0.375	2	1	1	0.125
P2	0.25	0.25	-0.625	-0.375	2	-1	1	0.125
P3	0.25	0.25	-0.125	-0.375	0.5	0.5	0.5	0.125
P4	0.25	0.25	-0.375	-0.625	2	1	-1	0.125
P5	0.25	0.25	0	0	0.125	0.25	0.25	0.125

Code:

```
import random
import math
def fitness_function(x):
    return x * math.sin(10 * math.pi * x) + 1

# 2. Initialize parameters
NUM_PARTICLES = 30
MAX_ITER = 100
W = 0.7
C1 = 1.5
C2 = 1.5
X_BOUND = (-1, 2)
V_MAX = 0.1

class Particle:
    def __init__(self):
        self.position = random.uniform(*X_BOUND)
        self.velocity = random.uniform(-V_MAX, V_MAX)
        self.best_position = self.position
        self.best_fitness = fitness_function(self.position)

    def update_velocity(self, global_best_position):
        r1 = random.random()
        r2 = random.random()
        cognitive = C1 * r1 * (self.best_position - self.position)
        social = C2 * r2 * (global_best_position - self.position)
        self.velocity = W * self.velocity + cognitive + social
        # Clamp velocity within limits
        self.velocity = max(min(self.velocity, V_MAX), -V_MAX)

    def update_position(self):
        self.position += self.velocity
        # Keep particle within bounds
        self.position = max(min(self.position, X_BOUND[1]), X_BOUND[0])

def particle_swarm_optimization():

    swarm = [Particle() for _ in range(NUM_PARTICLES)]

    global_best_particle = max(swarm, key=lambda p: p.best_fitness)
    global_best_position = global_best_particle.position
    global_best_fitness = global_best_particle.best_fitness

    for iteration in range(MAX_ITER):
        for particle in swarm:

            current_fitness = fitness_function(particle.position)

            if current_fitness > particle.best_fitness:
                particle.best_fitness = current_fitness
                particle.best_position = particle.position
```

```
            # Update global best
            if current_fitness > global_best_fitness:
                global_best_fitness = current_fitness
                global_best_position = particle.position

        for particle in swarm:
            particle.update_velocity(global_best_position)
            particle.update_position()


        print(f"Iteration {iteration+1:03d} | Best Fitness = {global_best_fitness:.5f} | Best X = {global_best_position:.5f}")

    print("\nOptimization complete!")
    print(f"Best solution: x = {global_best_position:.5f}, f(x) = {global_best_fitness:.5f}")

if __name__ == "__main__":
    particle_swarm_optimization()
```

Output:

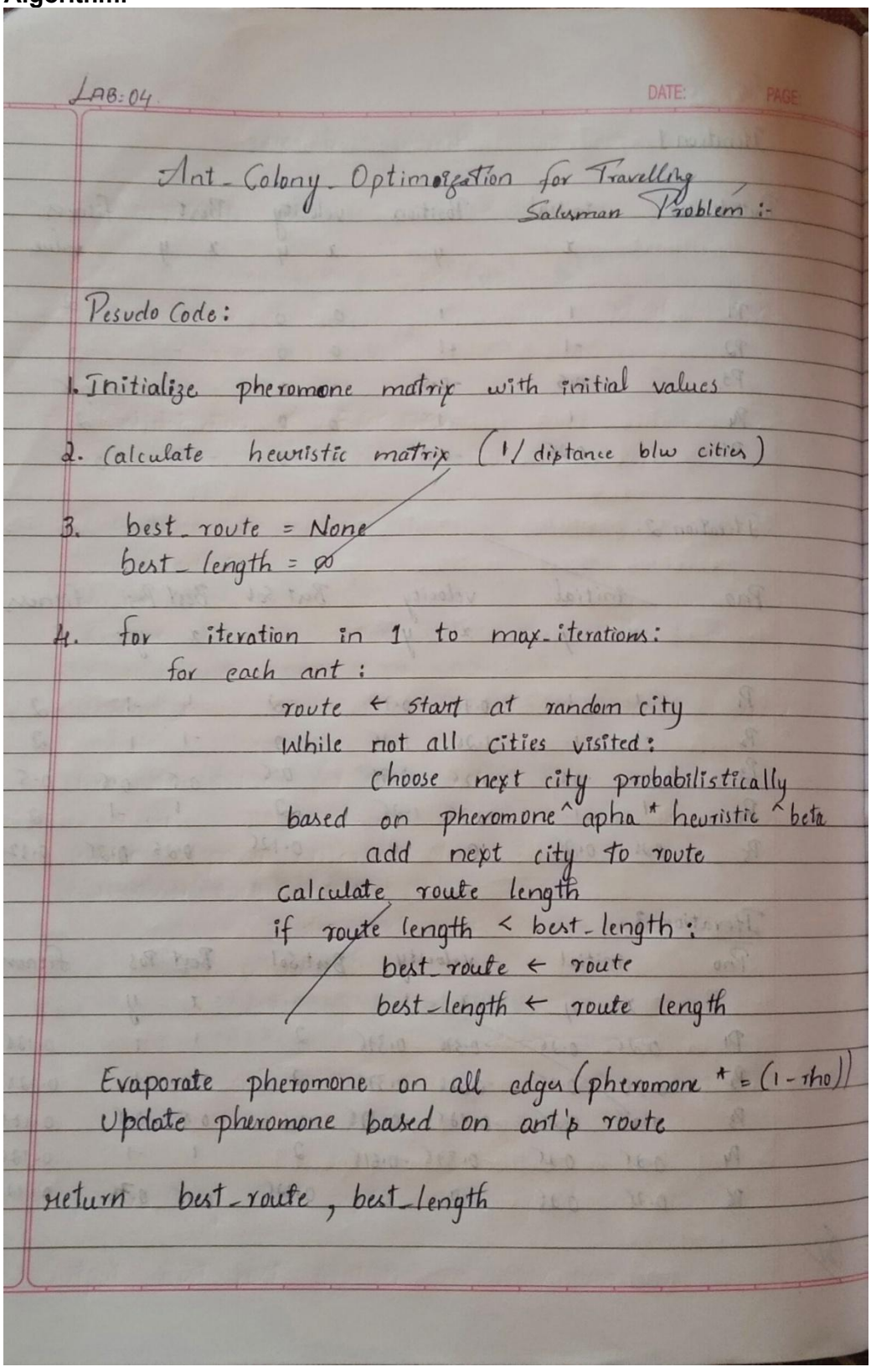
OUTPUT:

```
Iteration 1/20 - Best Fitness: 0.081095
Iteration 2/20 - Best Fitness: 0.081095
Iteration 3/20 - Best Fitness: 0.006725
Iteration 4/20 - Best Fitness: 0.003298
Iteration 5/20 - Best Fitness: 0.003298
Iteration 6/20 - Best Fitness: 0.003298
Iteration 7/20 - Best Fitness: 0.002956
Iteration 8/20 - Best Fitness: 0.002956
Iteration 9/20 - Best Fitness: 0.002956
Iteration 10/20 - Best Fitness: 0.002956
Iteration 11/20 - Best Fitness: 0.002956
Iteration 12/20 - Best Fitness: 0.001661
Iteration 13/20 - Best Fitness: 0.001066
Iteration 14/20 - Best Fitness: 0.001066
Iteration 15/20 - Best Fitness: 0.001066
Iteration 16/20 - Best Fitness: 0.000587
Iteration 17/20 - Best Fitness: 0.000587
Iteration 18/20 - Best Fitness: 0.000165
Iteration 19/20 - Best Fitness: 0.000095
Iteration 20/20 - Best Fitness: 0.000018

Best solution found:
Position: [ 0.00321441 -0.00268418]
Fitness: 1.7537291281392393e-05
```

Program 03: **Ant Colony Optimization for the Traveling Salesman Problem:**

Algorithm:



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Ant_Colony_Optimization for Travelling Salesman Problem :-

Pesudo Code:

1. Initialize pheromone matrix with initial values
2. Calculate heuristic matrix (1/distance blw cities)
3. best_route = None
   best_length = ∞
4. for iteration in 1 to max_iterations:
   for each ant:
   route ← start at random city
   While not all cities visited:
   choose next city probabilistically based on pheromone^apha * heuristic ^beta
   add next city to route
   Calculate route length
   if route length < best_length:
   best_route ← route
   best_length ← route length

Evaporate pheromone on all edges (pheromone *= (1-rho))
Update pheromone based on ant's route

return best_route, best_length

Output :-

Iteration 1/20 - Best length : 31.6195
Iteration 2/20 - Best length : 29.7691
:
:
Iteration 20/20 - Best length : 29.7691

Best route found : [3, 5, 6, 1, 0, 7, 2, 4]
Route length : 29.7691

Code:

```
import numpy as np

def distance(city1, city2):
    return np.linalg.norm(city1 - city2)

def route_length(route, cities):
    dist = 0.0
    for i in range(len(route) - 1):
        dist += distance(cities[route[i]], cities[route[i+1]])
    dist += distance(cities[route[-1]], cities[route[0]])
    return dist

def select_next_city(current_city, unvisited, pheromone, heuristic, alpha, beta):
    pheromone_vals = pheromone[current_city, unvisited] ** alpha
    heuristic_vals = heuristic[current_city, unvisited] ** beta
    probs = pheromone_vals * heuristic_vals
    probs /= probs.sum()
    return np.random.choice(unvisited, p=probs)


def aco_tsp(cities, num_ants=20, num_iterations=20, alpha=1.0, beta=5.0, rho=0.5,
initial_pheromone=1.0):
    num_cities = len(cities)


    pheromone = np.full((num_cities, num_cities), initial_pheromone)


    heuristic = np.zeros((num_cities, num_cities))
    for i in range(num_cities):
        for j in range(num_cities):
            if i != j:
                heuristic[i, j] = 1.0 / (distance(cities[i], cities[j]) + 1e-10)

    best_route = None
    best_length = float('inf')

    for iteration in range(num_iterations):
        all_routes = []
        all_lengths = []

        for ant in range(num_ants):
            route = []
            unvisited = list(range(num_cities))


            current_city = np.random.choice(unvisited)
            route.append(current_city)
            unvisited.remove(current_city)


            while unvisited:
                next_city = select_next_city(current_city, unvisited, pheromone,
heuristic, alpha, beta)
                route.append(next_city)
                unvisited.remove(next_city)
                current_city = next_city
```

```
            length = route_length(route, cities)
            all_routes.append(route)
            all_lengths.append(length)


            if length < best_length:
                best_length = length
                best_route = route

        pheromone *= (1 - rho)


        for route, length in zip(all_routes, all_lengths):
            for i in range(num_cities - 1):
                pheromone[route[i], route[i+1]] += 1.0 / length
                pheromone[route[i+1], route[i]] += 1.0 / length

            pheromone[route[-1], route[0]] += 1.0 / length
            pheromone[route[0], route[-1]] += 1.0 / length

        print(f"Iteration {iteration+1}/{num_iterations} - Best Length:
{best_length:.4f}")

    return best_route, best_length


if __name__ == "__main__":

    cities = np.array([
        [0, 0],
        [1, 5],
        [5, 2],
        [6, 6],
        [8, 3],
        [7, 9],
        [2, 8],
        [3, 3]
    ])

    best_route, best_length = aco_tsp(cities)

    print("\nBest route found:")
    print(best_route)
    print("Route length:", best_length)
```

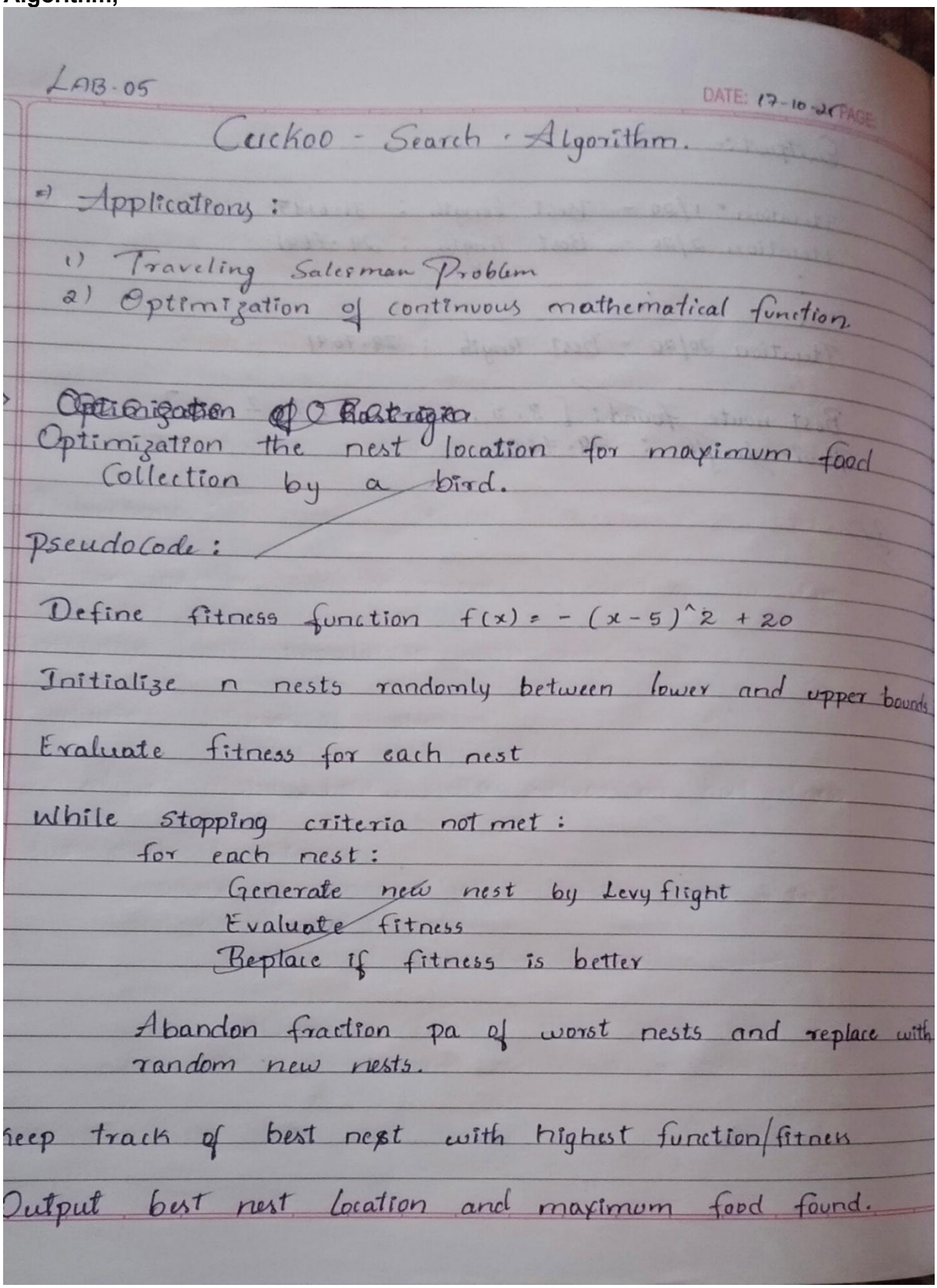
Output:

```
Iteration 1/20 - Best Length: 31.6195
Iteration 2/20 - Best Length: 29.7691
Iteration 3/20 - Best Length: 29.7691
Iteration 4/20 - Best Length: 29.7691
Iteration 5/20 - Best Length: 29.7691
Iteration 6/20 - Best Length: 29.7691
Iteration 7/20 - Best Length: 29.7691
Iteration 8/20 - Best Length: 29.7691
Iteration 9/20 - Best Length: 29.7691
Iteration 10/20 - Best Length: 29.7691
Iteration 11/20 - Best Length: 29.7691
Iteration 12/20 - Best Length: 29.7691
Iteration 13/20 - Best Length: 29.7691
Iteration 14/20 - Best Length: 29.7691
Iteration 15/20 - Best Length: 29.7691
Iteration 16/20 - Best Length: 29.7691
Iteration 17/20 - Best Length: 29.7691
Iteration 18/20 - Best Length: 29.7691
Iteration 19/20 - Best Length: 29.7691
Iteration 20/20 - Best Length: 29.7691

Best route found:
[np.int64(3), np.int64(5), np.int64(6), np.int64(1), np.int64(0), np.int64(7), np.int64(2), np.int64(4)]
Route length: 29.76913194777377
```

Program 04: **Cuckoo Search (CS):**

Algorithm;



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Cuckoo - Search - Algorithm.

=) Applications :

1) Traveling Salesman Problem
2) Optimization of continuous mathematical function

Optimization of ~~Cuckoo~~

Optimization the nest location for maximum food collection by a bird.

Pseudocode :

Define fitness function $f(x) = -(x-5)^2 + 20$

Initialize n nests randomly between lower and upper bounds

Evaluate fitness for each nest

While stopping criteria not met :
for each nest :
Generate new nest by Levy flight
Evaluate fitness
Replace if fitness is better

Abandon fraction pa of worst nests and replace with random new nests.

Keep track of best nest with highest function/fitness

Output best nest location and maximum food found.

Output:

Final Results:

=> Best nest Location : 5.0000

=> Maximum food Collected : 20.0000

Iterations:

Iteration 1 : Best nest Location = 5.3102
Max food = 19.9037

Iteration 20: Best nest Location = 4.9646
Max food = 19.9987

⋮

Iteration 80 : Best nest Location = 4.9992
Max food = 19.9999

⋮

Iteration 100 : Best nest Location = 5.000
Max food = 20.000

⋮

Iteration 150 : Best nest Location = 5.000
Max food = 20.000

Code:

```
import numpy as np
import math


def food_availability(x):
    return - (x - 5) ** 2 + 20

def levy_flight(Lambda=1.5):
    sigma_u = (math.gamma(1 + Lambda) * math.sin(math.pi * Lambda / 2) /
               (math.gamma((1 + Lambda) / 2) * Lambda * 2 ** ((Lambda - 1) / 2)))
** (1 / Lambda)
    sigma_v = 1
    u = np.random.normal(0, sigma_u)
    v = np.random.normal(0, sigma_v)
    step = u / (abs(v) ** (1 / Lambda))
    return step

def cuckoo_search_nest_location(n=10, iterations=200, pa=0.25, lower_bound=0,
upper_bound=10):
    nests = np.random.uniform(lower_bound, upper_bound, n)
    fitness = np.array([food_availability(x) for x in nests])

    best_idx = np.argmax(fitness)
    best_nest = nests[best_idx]
    best_fitness = fitness[best_idx]

    Lambda = 1.5

    for t in range(iterations):
        for i in range(n):
            step_size = 0.1 * levy_flight(Lambda)
            new_nest = nests[i] + step_size * (nests[i] - best_nest)
            new_nest = np.clip(new_nest, lower_bound, upper_bound)

            new_fitness = food_availability(new_nest)
            if new_fitness > fitness[i]:
                nests[i] = new_nest
                fitness[i] = new_fitness

                if new_fitness > best_fitness:
                    best_fitness = new_fitness
                    best_nest = new_nest

        num_abandon = int(pa * n)
        worst_indices = np.argsort(fitness)[:num_abandon]
        nests[worst_indices] = np.random.uniform(lower_bound, upper_bound,
num_abandon)
        fitness[worst_indices] = [food_availability(x) for x in
nests[worst_indices]]

        current_best_idx = np.argmax(fitness)
        if fitness[current_best_idx] > best_fitness:
            best_fitness = fitness[current_best_idx]
            best_nest = nests[current_best_idx]

        if (t+1) % 20 == 0 or t == 0:
            print(f"Iteration {t+1}: Best nest location = {best_nest:.4f}, Max
food = {best_fitness:.6f}")
```

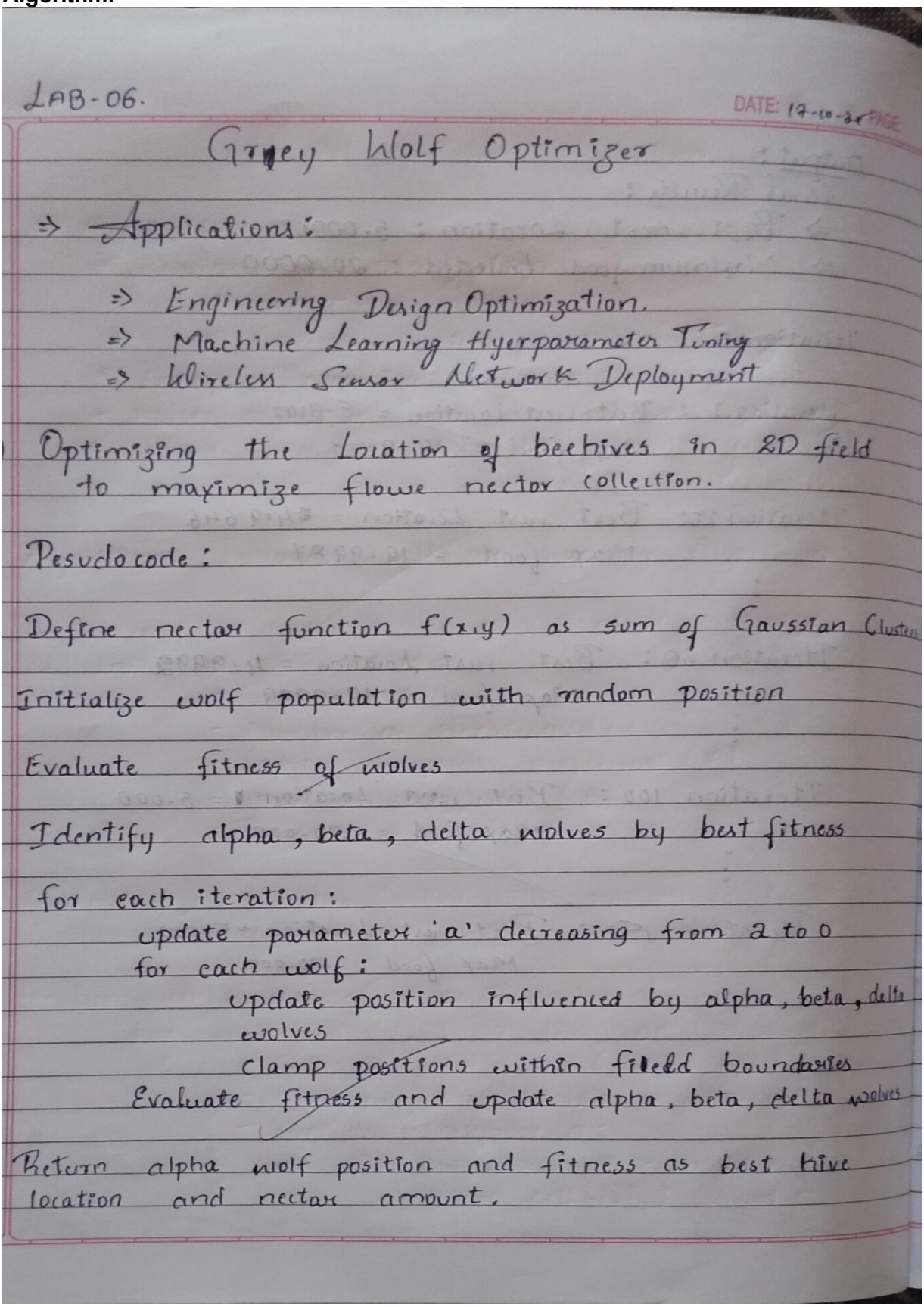
```
    return best_nest, best_fitness

if __name__ == "__main__":
    best_location, max_food = cuckoo_search_nest_location()
    print("\nFinal Results:")
    print(f"Best nest location: {best_location:.4f}")
    print(f"Maximum food collected: {max_food:.6f}")
```

Output:

```
Iteration 1: Best nest location = 5.3102, Max food = 19.903783
Iteration 20: Best nest location = 4.9646, Max food = 19.998744
Iteration 40: Best nest location = 4.9646, Max food = 19.998744
Iteration 60: Best nest location = 5.0038, Max food = 19.999985
Iteration 80: Best nest location = 4.9992, Max food = 19.999999
Iteration 100: Best nest location = 5.0000, Max food = 20.000000
Iteration 120: Best nest location = 5.0000, Max food = 20.000000
Iteration 140: Best nest location = 5.0000, Max food = 20.000000
Iteration 160: Best nest location = 5.0000, Max food = 20.000000
Iteration 180: Best nest location = 5.0000, Max food = 20.000000
Iteration 200: Best nest location = 5.0000, Max food = 20.000000

Final Results:
Best nest location: 5.0000
Maximum food collected: 20.000000
```

Program 05: **Grey Wolf Optimizer (GWO):**

Algorithm:



LAB-06. DATE: 17-10-25 PAGE

Grey Wolf Optimizer

=> Applications:

=> Engineering Design Optimization.
=> Machine Learning Hyerparameter Tuning
=> Wireless Sensor Network Deployment

Optimizing the location of beehives in 2D field to maximize flowe nector collection.

Pesudocode:

Define nector function f(x,y) as sum of Gaussian Clusters

Initialize wolf population with random position

Evaluate fitness of wolves

Identify alpha, beta, delta wolves by best fitness

for each iteration:
update parameter 'a' decreasing from 2 to 0
for each wolf:
update position influenced by alpha, beta, delta wolves
clamp positions within fileld boundaries
Evaluate fitness and update alpha, beta, delta wolves

Return alpha wolf position and fitness as best hive location and nectar amount.

Output :

Iteration 1 : Best nectar = 14.314 at location
$x = 5.9729$, $y = 6.9847$

⋮

Iteration 10 : Best nectar = 19.9269 at Location
$x = 3.0522$, $y = 3.0444$

⋮

Iteration 15 : Best nectar = 19.9769 at location
$x = 3.0522$, $y = 3.0444$

⋮

Iteration 20 : Best nectar = 19.9994 at Location
$x = 3.0134$, $y = 2.9978$

Final optimal beehive location :
$x = 3.0134$, $y = 2.9978$

Maximum nectar availability : 19.9994

Code:

```
import numpy as np

def nectar_availability(position):
    x, y = position
    term1 = 20 * np.exp(-((x - 3) ** 2 + (y - 3) ** 2) / 4)
    term2 = 15 * np.exp(-((x - 7) ** 2 + (y - 7) ** 2) / 3)
    term3 = 10 * np.exp(-((x - 5) ** 2 + (y - 8) ** 2) / 2)
    return term1 + term2 + term3

def gwo_beehive(n_wolves=30, iterations=20, lower_bound=0, upper_bound=10):
    dim = 2

    wolves = np.random.uniform(lower_bound, upper_bound, (n_wolves, dim))

    alpha_pos = np.zeros(dim)
    alpha_score = -np.inf
    beta_pos = np.zeros(dim)
    beta_score = -np.inf
    delta_pos = np.zeros(dim)
    delta_score = -np.inf

    for t in range(iterations):
        for i in range(n_wolves):
            fitness = nectar_availability(wolves[i])

            if fitness > alpha_score:
                alpha_score = fitness
                alpha_pos = wolves[i].copy()
            elif fitness > beta_score:
                beta_score = fitness
                beta_pos = wolves[i].copy()
            elif fitness > delta_score:
                delta_score = fitness
                delta_pos = wolves[i].copy()

        a = 2 - t * (2 / iterations)

        for i in range(n_wolves):
            for j in range(dim):
                r1 = np.random.rand()
                r2 = np.random.rand()
                A1 = 2 * a * r1 - a
                C1 = 2 * r2
                D_alpha = abs(C1 * alpha_pos[j] - wolves[i][j])
                X1 = alpha_pos[j] - A1 * D_alpha

                r1 = np.random.rand()
                r2 = np.random.rand()
                A2 = 2 * a * r1 - a
                C2 = 2 * r2
                D_beta = abs(C2 * beta_pos[j] - wolves[i][j])
                X2 = beta_pos[j] - A2 * D_beta

                r1 = np.random.rand()
                r2 = np.random.rand()
                A3 = 2 * a * r1 - a
                C3 = 2 * r2
                D_delta = abs(C3 * delta_pos[j] - wolves[i][j])
```

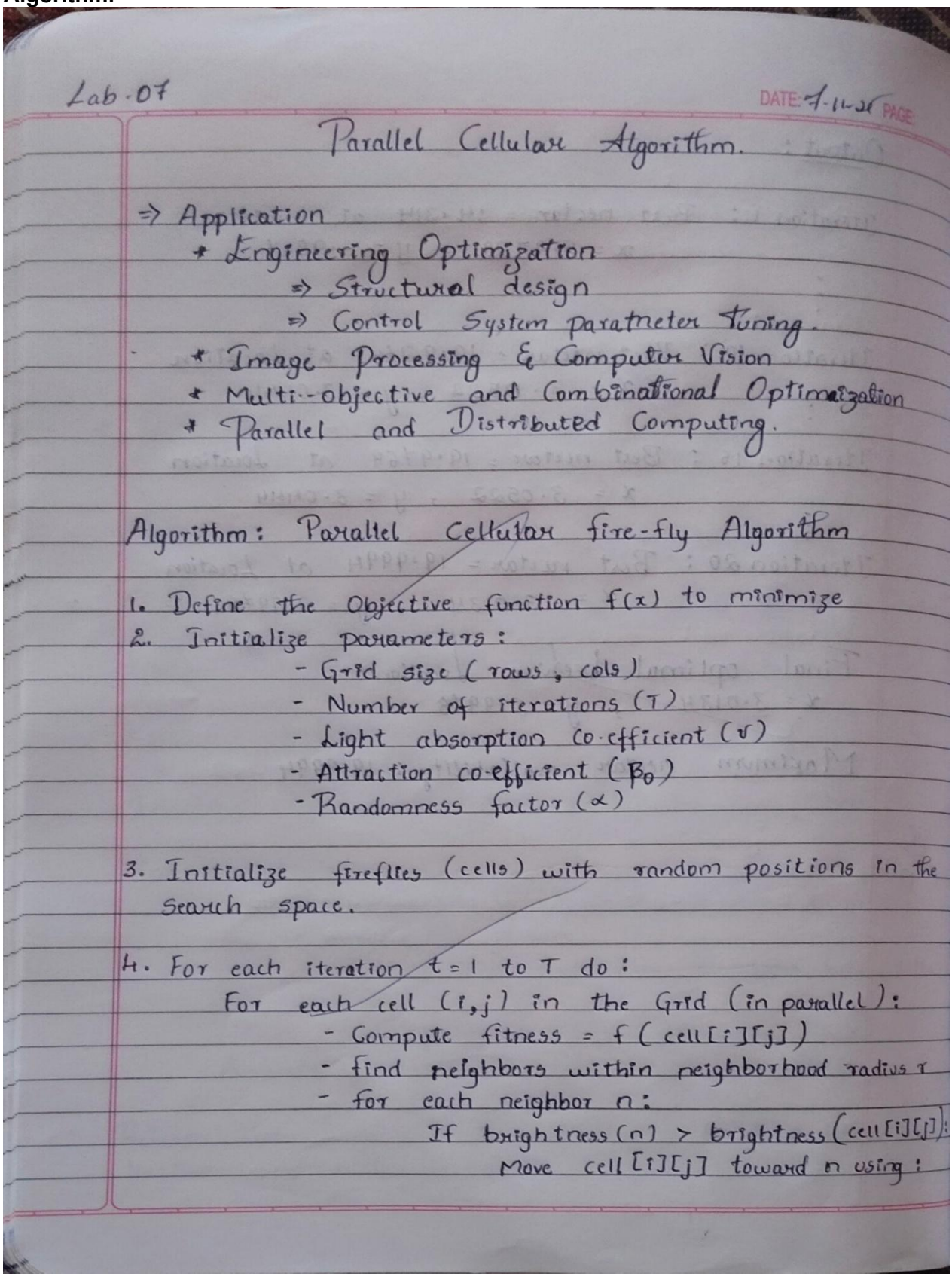
```
                X3 = delta_pos[j] - A3 * D_delta

                wolves[i][j] = (X1 + X2 + X3) / 3

            wolves[i] = np.clip(wolves[i], lower_bound, upper_bound)

        print(f"Iteration {t+1:2d}: Best nectar = {alpha_score:.6f} at location
x={alpha_pos[0]:.4f}, y={alpha_pos[1]:.4f}")

    return alpha_pos, alpha_score

if __name__ == "__main__":
    best_hive_location, max_nectar = gwo_beehive()
    print("\nFinal optimal beehive location:")
    print(f"x = {best_hive_location[0]:.4f}, y = {best_hive_location[1]:.4f}")
    print(f"Maximum nectar availability: {max_nectar:.6f}")
```

Output:

```
Iteration  1: Best nectar = 14.314430 at location x=5.9729, y=6.9847
Iteration  2: Best nectar = 14.314430 at location x=5.9729, y=6.9847
Iteration  3: Best nectar = 18.833972 at location x=3.4183, y=2.7443
Iteration  4: Best nectar = 18.833972 at location x=3.4183, y=2.7443
Iteration  5: Best nectar = 19.918544 at location x=2.8770, y=3.0355
Iteration  6: Best nectar = 19.918544 at location x=2.8770, y=3.0355
Iteration  7: Best nectar = 19.918544 at location x=2.8770, y=3.0355
Iteration  8: Best nectar = 19.918544 at location x=2.8770, y=3.0355
Iteration  9: Best nectar = 19.926097 at location x=3.1064, y=2.9403
Iteration 10: Best nectar = 19.976975 at location x=3.0522, y=3.0444
Iteration 11: Best nectar = 19.976975 at location x=3.0522, y=3.0444
Iteration 12: Best nectar = 19.976975 at location x=3.0522, y=3.0444
Iteration 13: Best nectar = 19.976975 at location x=3.0522, y=3.0444
Iteration 14: Best nectar = 19.976975 at location x=3.0522, y=3.0444
Iteration 15: Best nectar = 19.976975 at location x=3.0522, y=3.0444
Iteration 16: Best nectar = 19.976975 at location x=3.0522, y=3.0444
Iteration 17: Best nectar = 19.976975 at location x=3.0522, y=3.0444
Iteration 18: Best nectar = 19.976975 at location x=3.0522, y=3.0444
Iteration 19: Best nectar = 19.990873 at location x=2.9582, y=3.0120
Iteration 20: Best nectar = 19.999439 at location x=3.0134, y=2.9978

Final optimal beehive location:
x = 3.0134, y = 2.9978
Maximum nectar availability: 19.999439
```

Program 6: Parallel Cellular Algorithms and Programs:

Algorithm:



Lab-07

DATE: 7-11-25 PAGE

Parallel Cellular Algorithm.

⇒ Application
* Engineering Optimization
  ⇒ Structural design
  ⇒ Control System parameter Tuning.
* Image Processing & Computer Vision
* Multi-objective and Combinational Optimization
* Parallel and Distributed Computing.

Algorithm: Parallel Cellular fire-fly Algorithm

1. Define the Objective function f(x) to minimize
2. Initialize parameters:
   - Grid size (rows, cols)
   - Number of iterations (T)
   - Light absorption co-efficient (γ)
   - Attraction co-efficient (β_0)
   - Randomness factor (α)

3. Initialize fireflies (cells) with random positions in the search space.

4. For each iteration t = 1 to T do:
   For each cell (i,j) in the Grid (in parallel):
   - Compute fitness = f(cell[i][j])
   - find neighbors within neighborhood radius r
   - for each neighbor n:
     If brightness(n) > brightness(cell[i][j]):
     Move cell[i][j] toward n using:

continue :

position = position + β * (n - position) + α * rand_noise
where $\beta = \beta_0 * \exp(-\gamma * \text{distance}^2)$

- evaluate new fitness

End for

Track global best firefly and its fitness

5. Output the best solution found.

⇒ Output :

Iteration 0 : Best fitness = 0.113938
Iteration 20 : Best fitness = 0.000406
Iteration 40 : Best fitness = 0.000145
:
:
Iteration 180 : Best fitness = 0.00008.

Best Solution : [1.00253 1.00519]

Best fitness : 7.95067 ≈ 8.0

Code:

```
import numpy as np

def rosenbrock(x):
    return (1 - x[0])**2 + 100 * (x[1] - x[0]**2)**2

grid_size = (10, 10)
num_dimensions = 2
num_iterations = 200
beta0 = 1.0        # Base attractiveness
gamma = 1.0        # Light absorption coefficient
alpha = 0.2        # Randomness factor
radius = 1
search_bounds = (-2, 2)

cells = np.random.uniform(search_bounds[0], search_bounds[1],
                          size=(grid_size[0], grid_size[1], num_dimensions))

get neighbors (Moore neighborhood)
def get_neighbors(cells, i, j):
    neighbors = []
    for di in range(-radius, radius + 1):
        for dj in range(-radius, radius + 1):
            if di == 0 and dj == 0:
                continue
            ni, nj = (i + di) % grid_size[0], (j + dj) % grid_size[1]
            neighbors.append(cells[ni, nj])
    return np.array(neighbors)

best_solution = None
best_fitness = float('inf')

for t in range(num_iterations):
    new_cells = np.copy(cells)

    for i in range(grid_size[0]):
        for j in range(grid_size[1]):
            firefly = cells[i, j]
            fit_i = rosenbrock(firefly)
            neighbors = get_neighbors(cells, i, j)

            for n in neighbors:
                fit_n = rosenbrock(n)
                if fit_n < fit_i:  # brighter (better)
                    distance = np.linalg.norm(firefly - n)
                    beta = beta0 * np.exp(-gamma * distance**2)
                    step = beta * (n - firefly) + alpha * np.random.uniform(-1,
1, num_dimensions)
                    firefly = firefly + step
                    fit_i = rosenbrock(firefly)

            new_cells[i, j] = firefly

            if fit_i < best_fitness:
                best_fitness = fit_i
                best_solution = firefly

    cells = new_cells

    if t % 20 == 0:
```

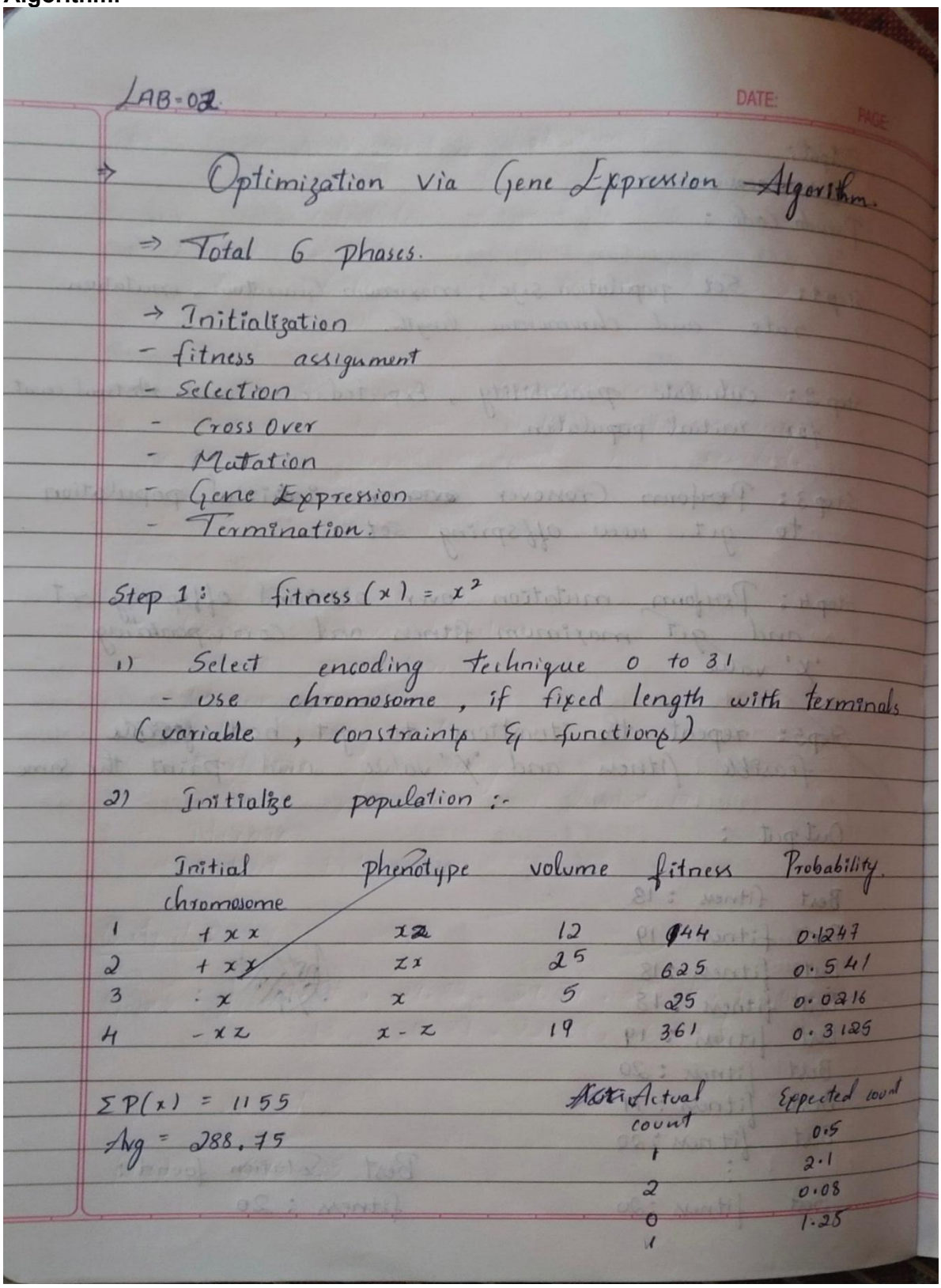
```
        print(f"Iteration {t}: Best Fitness = {best_fitness:.6f}")

print("\n============================")
print(" Parallel Cellular Firefly Algorithm Results")
print("============================")
print("Best Solution:", best_solution)
print("Best Fitness:", best_fitness)
```

Output:

```
Iteration 0: Best Fitness = 0.120322
Iteration 20: Best Fitness = 0.000162
Iteration 40: Best Fitness = 0.000114
Iteration 60: Best Fitness = 0.000049
Iteration 80: Best Fitness = 0.000049
Iteration 100: Best Fitness = 0.000049
Iteration 120: Best Fitness = 0.000049
Iteration 140: Best Fitness = 0.000049
Iteration 160: Best Fitness = 0.000009
Iteration 180: Best Fitness = 0.000009

============================
🔥 Parallel Cellular Firefly Algorithm Results
============================
Best Solution: [0.99738409 0.99491349]
Best Fitness: 8.760504339778715e-06
```

Program 7: Optimization via Gene Expression Algorithms:

Algorithm:



LAB-07 DATE: PAGE:

⇒ Optimization via Gene Expression Algorithm.

⇒ Total 6 phases.

→ Initialization
- fitness assignment
- Selection
- Cross Over
- Mutation
- Gene Expression
- Termination.

Step 1: fitness $(x) = x^2$

1) Select encoding technique 0 to 31
- use chromosome, if fixed length with terminals (variable, constraints & functions)

2) Initialize population :-

	Initial chromosome	phenotype	volume	fitness	Probability.
1	+ x x	x x	12	144	0.1247
2	+ x x	x x	25	625	0.5441
3	. x	x	5	25	0.0216
4	- x z	x - z	19	361	0.3125

$\Sigma P(x) = 1155$

Avg = 288.75

Actual count	Expected count
1	0.5
2	2.1
0	0.08
1	1.25

3. Selection of mating pool.

	Selected chromosome	Crossover pool	offspring	phenotype	value
1	+xx	2	+xx	x(x+1)	13
2	+xx	1	4xx	2x	24
3	+xx	3	+x-	x(x-)	27
4	-xz	1	+xz	x+z	17

fitness.
169
576
729.
289

4. Crossover : perform crossover randomly choosen gene position
max fitness after crossover = 729

5 Mutation :

	offspring after mutation	mutation applied	offspring after mutation	phenotype	x value
1	+x+	+ → -	*x-	x*(x-)	29
2	+xx	None	+xx	2x	24
3	+x-	- → x	+xx	x-	27
4	+xz	none	+xz	x+z	20

fitness
841
576
729
400

6) Gene Expression & Evolution:
Decode each genotype → phenotype
calculate fitness.

$\Sigma P(x) = 2546$
Avg = 636.5.
Max = 341

7) Iterate until ~~threshold~~ convergence
repeat step 3 – 6 until fitness improvement
negligible / generation limit has reached.

Pseudo code

- Define fitness Evolution
- Create population.
- Define parameters
- Select mating pool.
- Mutation after mating.
- Gene expression and Evolution.
- Iterate.
- Output Best value.

Output:

Genes : [29.53 , 29.82 , 29.34 , 28.54 , 15.01 , 21.08 ,
23.13 , 30.81 , 22.51 , 26.22]

$x = 26.37$

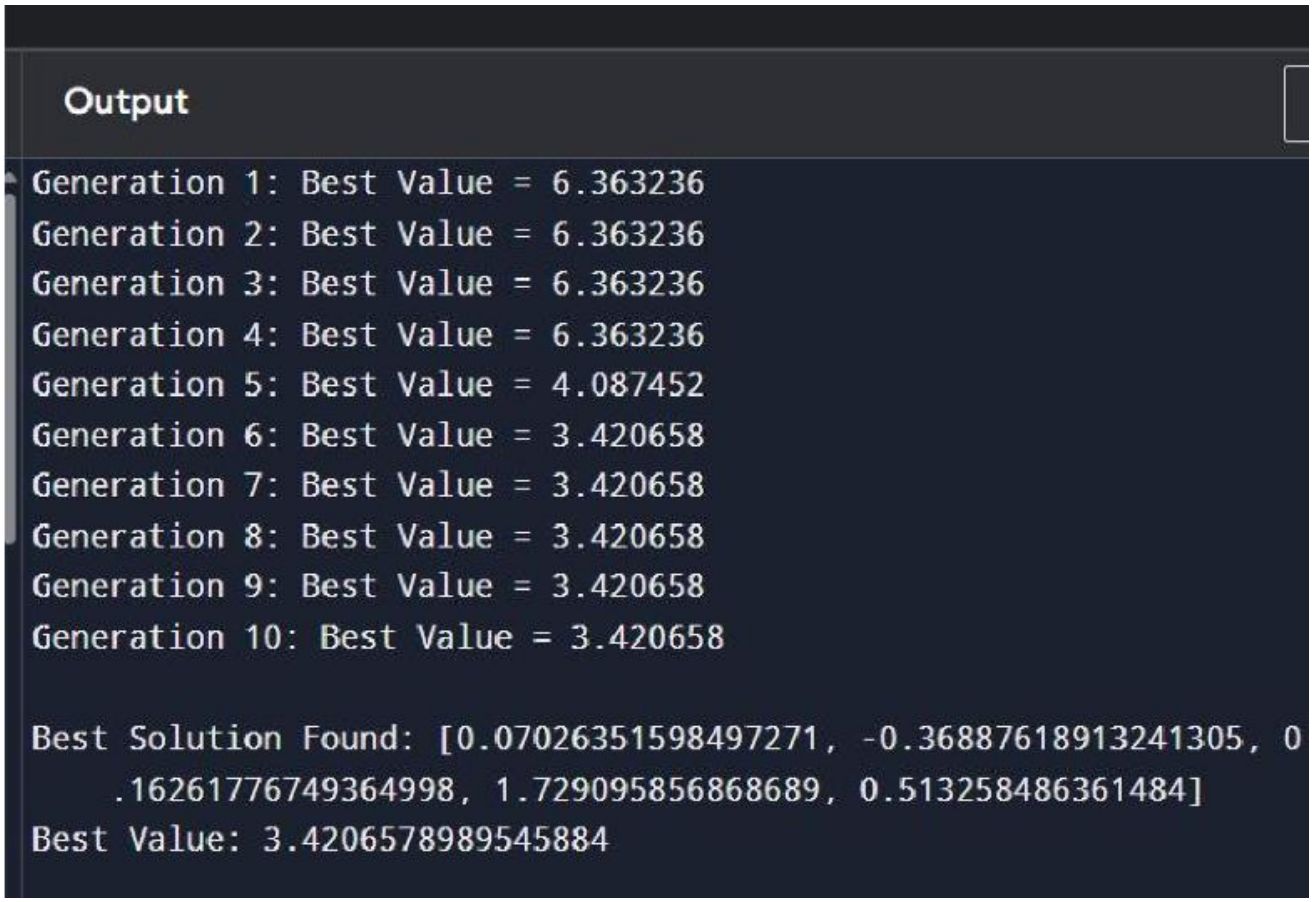
$f(x) = 695.45$.

Code:

```
import random
import math

def fitness_function(x):
    return x * math.sin(10 * math.pi * x) + 1  # Objective: maximize this

POPULATION_SIZE = 30
NUM_GENES = 10          # Number of genes per chromosome
MUTATION_RATE = 0.1
CROSSOVER_RATE = 0.8
GENERATIONS = 100
X_BOUND = (-1, 2)

def create_individual():
    return [random.uniform(X_BOUND[0], X_BOUND[1]) for _ in range(NUM_GENES)]

def create_population(size):
    return [create_individual() for _ in range(size)]

def express_genes(individual):
    # Expression step: translate gene sequence into a single functional value (x)
    expressed_x = sum(individual) / len(individual)
    return expressed_x

def evaluate_fitness(individual):
    x = express_genes(individual)
    return fitness_function(x)

def tournament_selection(population, k=3):
    selected = random.sample(population, k)
    selected.sort(key=lambda ind: evaluate_fitness(ind), reverse=True)
    return selected[0]

def crossover(parent1, parent2):
    if random.random() < CROSSOVER_RATE:
        point = random.randint(1, NUM_GENES - 1)
        child1 = parent1[:point] + parent2[point:]
        child2 = parent2[:point] + parent1[point:]
        return child1, child2
    else:
        return parent1[:], parent2[:]

def mutate(individual):
    for i in range(NUM_GENES):
        if random.random() < MUTATION_RATE:
            individual[i] += random.uniform(-0.1, 0.1)
            individual[i] = max(min(individual[i], X_BOUND[1]), X_BOUND[0])
    return individual
def gene_expression_algorithm():
    population = create_population(POPULATION_SIZE)
    best_individual = None
    best_fitness = float('-inf')
```

```
    for generation in range(GENERATIONS):
        new_population = []


        for ind in population:
            fit = evaluate_fitness(ind)
            if fit > best_fitness:
                best_fitness = fit
                best_individual = ind

        while len(new_population) < POPULATION_SIZE:
            parent1 = tournament_selection(population)
            parent2 = tournament_selection(population)
            child1, child2 = crossover(parent1, parent2)
            child1 = mutate(child1)
            child2 = mutate(child2)
            new_population.extend([child1, child2])

        population = new_population[:POPULATION_SIZE]

        expressed_x = express_genes(best_individual)
        print(f"Generation {generation+1:03d} | Best Fitness = {best_fitness:.5f} | Best X = {expressed_x:.5f}")

    final_x = express_genes(best_individual)
    print("\nOptimization complete!")
    print(f"Best solution: x = {final_x:.5f}, f(x) = {best_fitness:.5f}")

if __name__ == "__main__":
    gene_expression_algorithm()
```

Output:


Output
Generation 1: Best Value = 6.363236
Generation 2: Best Value = 6.363236
Generation 3: Best Value = 6.363236
Generation 4: Best Value = 6.363236
Generation 5: Best Value = 4.087452
Generation 6: Best Value = 3.420658
Generation 7: Best Value = 3.420658
Generation 8: Best Value = 3.420658
Generation 9: Best Value = 3.420658
Generation 10: Best Value = 3.420658
Best Solution Found: [0.07026351598497271, -0.36887618913241305, 0
.16261776749364998, 1.729095856868689, 0.513258486361484]
Best Value: 3.4206578989545884